

## A: Project title

Aligning Multi-Agent Interactions for Sustainable Food Behavior (AMAI)

## B: In a nutshell

The project addresses a critical sustainability challenge: reducing household food waste. While existing Artificial Intelligence (AI) systems are typically optimized based on shallow user preferences, I propose an innovative paradigm shift by aligning user behavior directly with long-term sustainable goals. I will develop a multi-agent system where user-facing AI agents (powered by large language models, LLMs) dynamically interact with computational agents capable of performing complex environmental impact assessments using external databases and calculators. For example, the system might detect high-emission food items (e.g., minced beef nearing expiration) and proactively suggest practical solutions, accompanied by narratives explaining the environmental importance and guidance for reducing waste effectively, including personalized recipes. I will demonstrate a groundbreaking method for AI alignment, where measurable sustainable outcomes, rather than just user-stated preferences, guide AI interactions—transforming consumer behavior in sustainability contexts beyond food waste alone.

## C: Research idea and context

Reducing food waste is critical to addressing climate change, as food production significantly contributes to global greenhouse gas emissions [2]. Despite widespread awareness and numerous interventions, household-level food waste remains stubbornly high, indicating substantial limitations in existing solutions [4]. Current consumer-oriented strategies predominantly rely on providing information or generic recommendations, often neglecting personalized context and the individual’s likelihood of adherence. Recent advances in AI, particularly LLMs, have significantly improved personalized interaction capabilities, yet traditional alignment methods primarily optimize immediate conversational satisfaction based on user preferences [3], failing to guarantee alignment with long-term goals. A crucial open research question, therefore, is whether AI can effectively align user behavior to sustainability goals. This project addresses this gap.

I build on the state-of-the-art in goal-oriented reinforcement learning [1] and multi-agent systems that can use external tools [11]. While AI alignment research typically employs reinforcement learning from human feedback to optimize user satisfaction [7], I introduce a pioneering approach where agents actively use computational tools to calculate environmental impacts and optimize behavioral outcomes. These multi-agent systems will dynamically consider factors such as food inventory, expiration dates, environmental impact metrics, and anticipated behavioral compliance, making recommendations that balance

both feasibility and impact. Additionally, I integrate recent insights from computational argumentation research [10, 8] and recipe adaptation methods [5, 6], where narrative generation and personalized content are used to enhance user engagement and adherence. By integrating agent-based computational models with actionable recommendations, I aim to demonstrate for the first time how explicit sustainable behavior goals, rather than short-term user feedback alone, can effectively guide AI alignment strategies in practical scenarios.

## **D: Proposed method of solution or concept**

A two-year postdoc will develop and validate a novel multi-agent AI framework: LLM-driven agents will generate personalized recipes and narratives tailored to users' inventories and preferences; and calculate environmental impacts and predict user adherence to recommendations, leveraging external tools and databases [9]. These agents are jointly optimized based on behavioral predictions and quantified sustainability impact, focusing especially on reducing waste of high-impact food items (e.g., beef, fish, dairy).

A mobile app will be developed for this project and serve as the main platform for interaction. 20–30 participants, recruited with informed consent, will use the app daily for one year. A one-year student assistant will manage data collection, participant communication, and logistical aspects of the user study. Throughout the study, we will measure behavior changes (e.g., food waste reduction using kitchen diaries) and self-reported goal attainment.

Two one-month international research stays by the postdoc will strengthen methodological rigor: the first on recipe modeling and adaptation, ensuring accurate and coherent recommendations, and the second on computational argumentation to craft persuasive narratives supporting sustainable behaviors.

Evaluation metrics and success criteria include quantifiable reductions in household food waste, carbon emissions, and improved alignment of user behavior with explicitly stated personal goals (such as environmental impact reduction, ethical eating, or enjoyment of meals). Validation through surveys will confirm users' perceived achievement of their individual objectives.

## **E: Major gains and obstacles**

The novel multi-agent AI framework will extend the concept of alignment to goal-oriented interactions. By demonstrating the feasibility of integrating multi-agent LLM systems with external tools, I will advance AI alignment, providing a foundation for future AI systems aimed at achieving measurable, socially beneficial outcomes.

Obstacles include accurately predicting user compliance, handling complex integrations with computational tools, and effectively training reinforcement learning models, which is computationally challenging due to the delayed nature of behavioral outcomes used as rewards.

## F: Appropriateness

The ambitious integration of tool-enabled multi-agent delayed-reward reinforcement learning with user studies and sustainability represents a novel interdisciplinary approach with considerable uncertainty. Conventional funding typically favors incremental innovation and established methodologies, whereas this project deliberately embraces uncertainty to explore potentially transformative AI alignment capabilities.

## G: Probable objections

Objections could be raised about influencing user behavior, and whether shifting from preference-based to goal-oriented alignment might negatively affect user autonomy by nudging humans toward specific behaviors. I acknowledge these ethical considerations and explicitly address them by emphasizing informed consent, transparency about AI recommendations, and the explicit alignment of the system’s goals with user-defined objectives. An objection could be raised regarding the practical challenges of integrating multi-agent systems with computational tools, given the complexity of coordinating different models and APIs in real-time. This concern will be addressed through iterative development and continuous testing within controlled user studies, allowing incremental refinement and validation.

## References

- [1] Liyu Chen and Haipeng Luo. Near-optimal goal-oriented reinforcement learning in non-stationary environments, 2022.
- [2] Monica Crippa, Efsio Solazzo, Diego Guizzardi, Fabio Monforti-Ferrario, Francesco N. Tubiello, and Adrian Leip. Food systems are responsible for a third of global anthropogenic GHG emissions. *Nature Food*, 2(3):198–209, 2021.
- [3] Hannah Rose Kirk, Alexander Whitefield, Paul Rottger, Andrew M. Bean, Katerina Margatina, Rafael Mosquera-Gomez, Juan Ciro, Max Bartolo, Adina Williams, He He, Bertie Vidgen, and Scott Hale. The PRISM alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 105236–105344. Curran Associates, Inc., 2024.
- [4] Ana Carolina B Leme, Sophia Hou, Regina Mara Fisberg, Mauro Fisberg, and Jess Haines. Adherence to food-based dietary guidelines: a systemic

- review of high-income and low-and middle-income countries. *Nutrients*, 13(3):1038, 2021.
- [5] Xiao Liu, Yansong Feng, Jizhi Tang, Chengang Hu, and Dongyan Zhao. Counterfactual recipe generation: Exploring compositional generalization in a realistic scenario. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 7354–7370, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics.
  - [6] A. Morales-Garzón, K. Gutiérrez-Batista, and M. J. Martín-Bautista. Adaptafood: An intelligent system to adapt recipes to specialized dietary constraints. *Multimedia Systems*, 2025.
  - [7] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
  - [8] Amalie Brogaard Pauli, Isabelle Augenstein, and Ira Assent. Measuring and benchmarking large language models’ capabilities to generate persuasive language. In *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics.
  - [9] Joseph Poore and Thomas Nemecek. Reducing food’s environmental impacts through producers and consumers. *Science*, 360(6392):987–992, 2018.
  - [10] Francesco Salvi, Manoel Horta Ribeiro, Riccardo Gallotti, and Robert West. On the conversational persuasiveness of large language models: A randomized controlled trial, 2024.
  - [11] Weizhou Shen, Chenliang Li, Hongzhan Chen, Ming Yan, Xiaojun Quan, Hehong Chen, Ji Zhang, and Fei Huang. Small LLMs are weak tool learners: A multi-LLM agent. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 16658–16680, Miami, Florida, USA, November 2024. Association for Computational Linguistics.